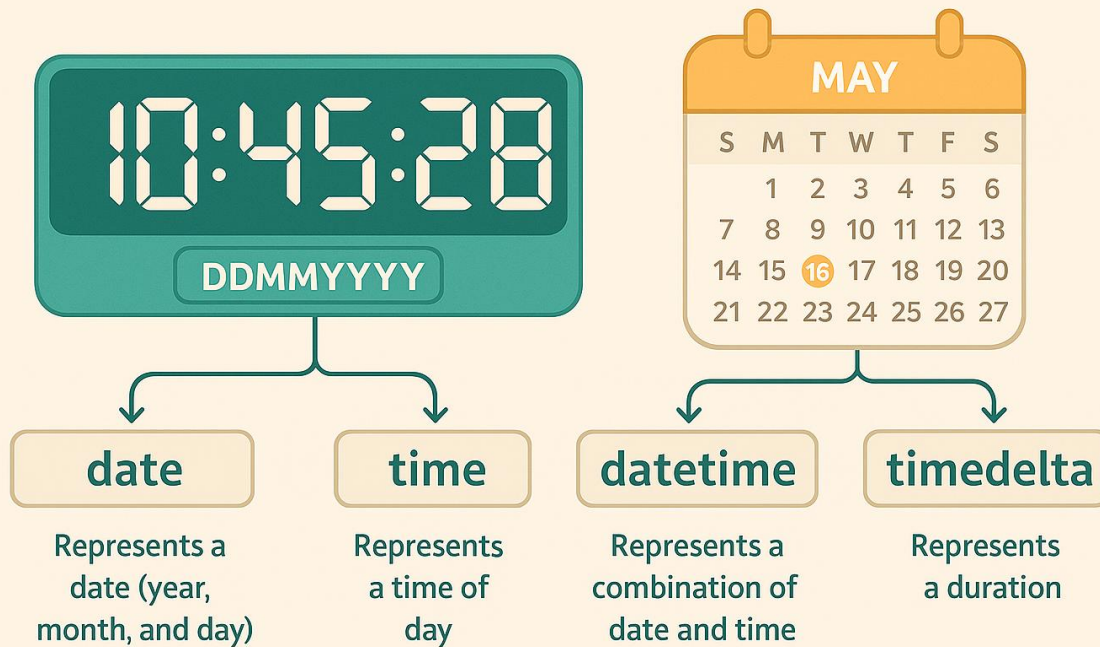


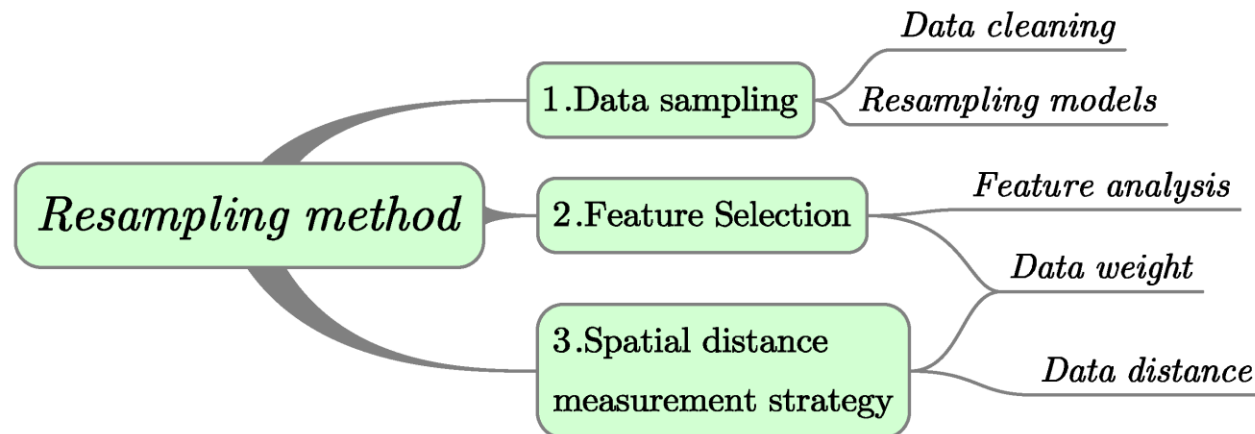
# THE DIMENSION OF TIME



- **Definition:** A sequence of data points indexed in successive order, usually with a fixed frequency (daily, monthly, yearly).
- **The Temporal Engine:** In Python, we move from standard indices to DatetimeIndex, which allows us to treat time as a first-class mathematical object.
- **Context:** Analyzing rainfall in Swat or price fluctuations in Karkhano Market requires looking at trends over years, not just isolated points.

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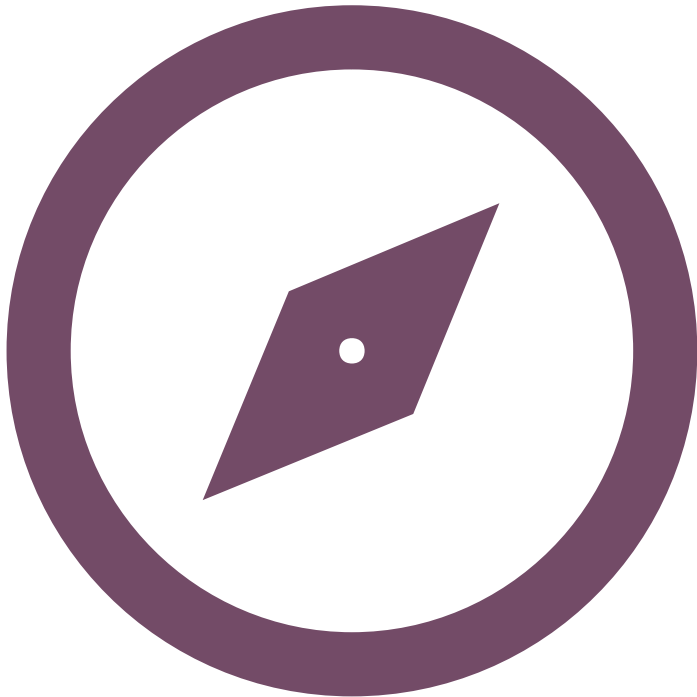
# RESAMPLING – CHANGING THE LENS



- **Downsampling:** Aggregating high-frequency data into lower frequency (e.g., Daily → Monthly) to find the "Big Picture".
  - **Upsampling:** Increasing frequency (e.g., Quarterly → Monthly) and "filling in" the gaps using interpolation.
  - **The Purpose:** Removing daily "noise" to reveal underlying quarterly or annual trends.
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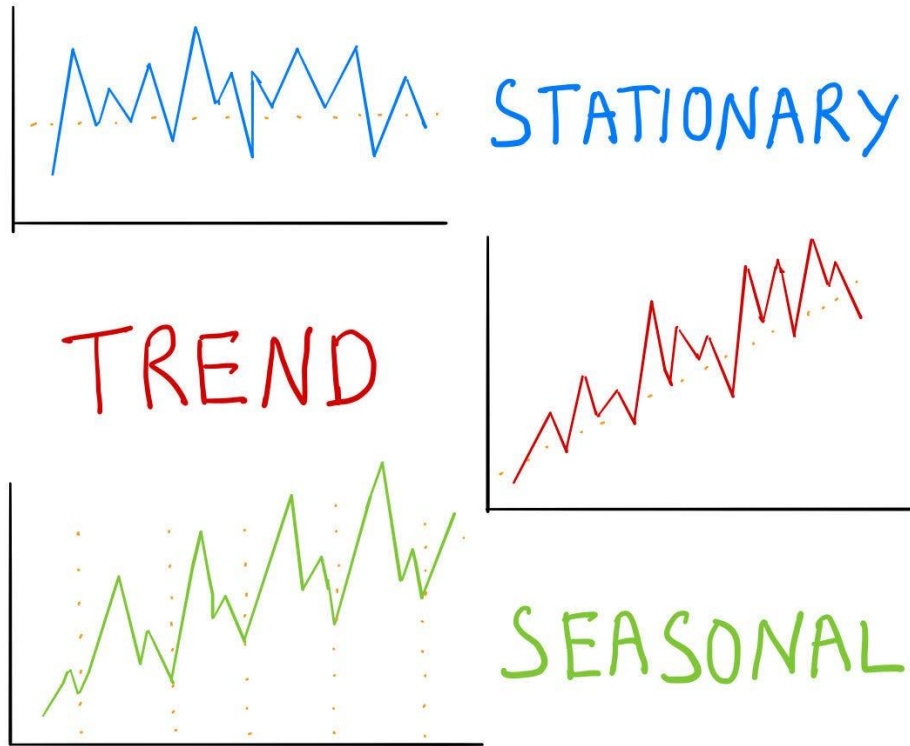
# LAG ANALYSIS – THE MEMORY OF DATA



- **The Concept:** Using past values to explain current ones.
  - **Mechanical Necessity:** Many regional phenomena have a delay. For example, heavy rain in March (Lag) affects the crop yield in June.
  - **Autocorrelation:** Measuring how a variable is correlated with a delayed version of itself.
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# STATIONARITY VS. TRENDS



- **Trends:** A long-term increase or decrease in the data.
  - **Stationarity:** A property where the mean and variance do not change over time. Most statistical models require data to be "Stationary" to make valid predictions.
  - **Detrending:** The process of removing a trend (e.g., through "Differencing") to make the data stationary.
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